Amazon VPC — Important Interview Questions & Answers

1. What is Amazon VPC?
Answer:
Amazon VPC stands for **Amazon Virtual Private Cloud**. It is an AWS service used to create a private network for our AWS resources. Inside a VPC, we can control where our resources are placed and how they communicate with each other. We can define IP ranges, subnets, route tables, internet access, and network security.
For example, suppose we have an application running on EC2 and a database running on RDS. We can place both inside a VPC and control which resources can communicate with each other.
Example:
AWS
↓
VPC
├— EC2
└— RDS
Simple ga gurthupettuko:
VPC = **Private AWS Network + IP Range + Subnets + Routing + Security**

---

2. Why do we use VPC?
Answer:
We use VPC to create and control the network environment for our AWS resources.
It allows us to decide which resources should be public, which should be private, how traffic should move, and who can access our resources.
For example, we can keep a Load Balancer in a public subnet and keep our application servers and database in private subnets.
Example:
Internet
↓
Load Balancer
↓
Public Subnet
↓
Private Subnet
↓
EC2 / RDS
Simple ga gurthupettuko:
VPC = **Control our AWS network**

---

3. What is a VPC CIDR?
Answer:
A VPC CIDR is the IP address range that we assign to our VPC.
It defines which private IP addresses can be used inside the VPC. We choose the CIDR based on the number of resources and subnets we need.
For example, if we create a VPC with 10.0.0.0/16, the VPC gets a large private IP range that can be divided into smaller subnet ranges.
Example:
VPC
CIDR: 10.0.0.0/16

↓
Can create smaller subnets
Simple ga gurthupettuko:
CIDR = **VPC IP address range**

---

4. What is a Subnet?
Answer:
A subnet is a smaller network inside a VPC.
We divide the VPC IP range into different subnets so that we can organize our resources and control their network access.
A subnet belongs to one Availability Zone.
For example, we can create one public subnet for a Load Balancer and private subnets for EC2 and RDS.
Example:
VPC
|
|-- Public Subnet
|
|-- Private Subnet
|
|-- Private Subnet
Simple ga gurthupettuko:
Subnet = **Small network inside VPC**

---

5. What is the difference between Public Subnet and Private Subnet?

Answer:
A public subnet is a subnet that has a route to an Internet Gateway, so resources in that subnet can have internet connectivity when their configuration allows it.
A private subnet does not have a direct route to an Internet Gateway. It is commonly used for resources that should not be directly reachable from the internet.
For example, a Load Balancer can be placed in public subnets, while application servers and databases can be placed in private subnets.
Example:
Internet
↓
Public Subnet
↓
Load Balancer
↓
Private Subnet
↓
EC2 / RDS
Simple ga gurthupettuko:
Public Subnet = **Internet-facing**
Private Subnet = **Not directly internet-facing**

---

6. What is an Availability Zone in VPC?
Answer:
An Availability Zone, or AZ, is a separate location inside an AWS Region.
We can create subnets in different Availability Zones to improve application availability. A subnet belongs to only one AZ.
For example, we can create one subnet in AZ-1 and another subnet in AZ-2, and place application resources across both.
Example:
Region
|
|-- AZ-1
| ↓
| Subnet-A
|
|-- AZ-2
↓
Subnet-B
Simple ga gurthupettuko:
AZ = **Separate location inside a Region**

---

7. What is an Internet Gateway?
Answer:
An Internet Gateway, or IGW, allows communication between a VPC and the internet.
It is attached to the VPC, and a subnet needs a route to the Internet Gateway to be considered public.
For example, if an EC2 instance in a public subnet needs to communicate directly with the internet, the VPC needs an Internet Gateway and the subnet route table must have the correct route.
Example:
Internet
↓
Internet Gateway
↓
VPC
↓
Public Subnet
↓
EC2
Simple ga gurthupettuko:
Internet Gateway = **VPC ↔ Internet connection**

---

8. What is a Route Table?
Answer:
A route table contains rules that decide where network traffic should go.
Each route has a destination and a target. Subnets use route tables to decide how traffic should be forwarded.
For example, a public subnet can have a route such as 0.0.0.0/0 pointing to an Internet Gateway.
Example:
Public Subnet
↓
Route Table
↓
0.0.0.0/0
↓
Internet Gateway

↓
Internet
Simple ga gurthupettuko:
Route Table = **Traffic ekkadiki vellalo decide chestundi**

---

9. What is 0.0.0.0/0?
Answer:
0.0.0.0/0 represents all IPv4 addresses.
In a route table, it is commonly used as the default route, meaning traffic for destinations that do not match a more specific route should follow this route.
For example, if we have:
0.0.0.0/0 → Internet Gateway
it means internet-bound IPv4 traffic can be sent through the Internet Gateway, subject to other network configuration.
Example:
EC2
↓
Route Table
↓
0.0.0.0/0
↓
Internet Gateway
↓
Internet
Simple ga gurthupettuko:
0.0.0.0/0 = **All IPv4 destinations**

---

10. What is a NAT Gateway?
Answer:
A NAT Gateway allows resources in a private subnet to connect to the internet for outbound traffic, without allowing the internet to directly start a connection to those private resources.
It is commonly used when private servers need to download updates, install packages, or access external services.
For example, an EC2 instance in a private subnet can use a NAT Gateway to download software updates from the internet.
Example:
Private EC2
↓
Route Table
↓
NAT Gateway
↓
Internet Gateway
↓
Internet
Simple ga gurthupettuko:
NAT Gateway = **Private resources → Internet outbound access**

---

11. What is the difference between Internet Gateway and NAT Gateway?
Answer:
An Internet Gateway provides internet connectivity for resources in a VPC when the routing and resource configuration allow it.
A NAT Gateway is mainly used to give resources in private subnets outbound internet access without making those private resources directly reachable from the internet.
For example, a public EC2 can use an Internet Gateway, while a private EC2 can use a NAT Gateway for outbound internet access.
Example:
Public EC2
↓
Internet Gateway
↓
Internet

Private EC2
↓
NAT Gateway
↓
Internet Gateway
↓
Internet
Simple ga gurthupettuko:
IGW = **Internet connectivity**
NAT Gateway = **Private subnet outbound internet**

---

12. What is a Security Group?
Answer:

A Security Group is a virtual firewall attached to resources such as EC2 instances.
It controls inbound and outbound traffic based on rules such as protocol, port, and source or destination.
Security Groups are **stateful**, which means return traffic is automatically allowed for an allowed connection.
For example, we can allow HTTP traffic on port 80 from the Load Balancer to an EC2 instance.
Example:
Load Balancer
↓
Security Group
↓
Port 80 Allowed
↓
EC2
Simple ga gurthupettuko:
Security Group = **Resource-level firewall**

---

13. What is a Network ACL?
Answer:
A Network ACL, or NACL, is a network-level firewall associated with a subnet.
It controls inbound and outbound traffic using numbered rules. NACLs are **stateless**, so inbound and outbound traffic must be allowed separately when required.
For example, we can use a NACL to allow or deny traffic at the subnet level.
Example:
VPC
↓
Subnet
↓
NACL
↓
Resources
Simple ga gurthupettuko:
NACL = **Subnet-level firewall**

---

14. What is the difference between Security Group and NACL?
Answer:
A Security Group works at the resource level, while a NACL works at the subnet level.
Security Groups are stateful, while NACLs are stateless and use numbered rules that are evaluated in order.
For example, an EC2 instance can have a Security Group, while the subnet containing that EC2 can have a NACL.
Example:
VPC
↓
NACL
↓
Subnet
↓
Security Group
↓
EC2
Simple ga gurthupettuko:
Security Group = **Resource level**
NACL = **Subnet level**

---

15. What is a private IP address in VPC?
Answer:
A private IP address is an IP address used for communication inside private networks.
AWS resources inside a VPC commonly communicate with each other using private IP addresses.
Private IP addresses are not directly reachable from the public internet.
For example, an EC2 instance may have a private IP such as 10.0.1.10 and communicate with an RDS database using private networking.
Example:
EC2
10.0.1.10
↓
Private Network
↓
RDS
10.0.2.10
Simple ga gurthupettuko:
Private IP = **Internal network communication**

---

16. What is a Public IP address?
Answer:

A public IP address is an IP address that can be used for communication with the internet.
A resource needs the correct network configuration, including routing and security rules, to communicate through the internet.
For example, an EC2 instance in a public subnet can have a public IPv4 address and use an Internet Gateway for internet communication.
Example:
Internet
↓
Public IP
↓
EC2
Simple ga gurthupettuko:
Public IP = **Internet-facing IP**

---

17. What is an Elastic IP?
Answer:
An Elastic IP is a static public IPv4 address that we can allocate to our AWS account and associate with supported resources.
Unlike a normal public IP that can change in some situations, an Elastic IP remains associated with our account until we release it.
For example, if an application needs a fixed public IPv4 address, we can use an Elastic IP.
Example:
Elastic IP
↓
EC2
↓
Application
Simple ga gurthupettuko:
Elastic IP = **Static public IPv4 address**

---

18. What is VPC Peering?
Answer:
VPC Peering allows two VPCs to communicate with each other using private IP addresses.
The VPCs can be in the same Region or, depending on the supported setup, different Regions.
The required route table and security rules must also be configured.
For example, one VPC may contain an application and another VPC may contain shared services. VPC Peering can allow private communication between them.
Example:
VPC-A
↓
VPC Peering
↓
VPC-B
Simple ga gurthupettuko:
VPC Peering = **Private connection between two VPCs**

---

19. What is a VPC Endpoint?
Answer:
A VPC Endpoint allows resources inside a VPC to access supported AWS services without sending the traffic through the public internet.
This is useful for keeping AWS service communication inside the AWS network.
There are different types of VPC endpoints, such as Gateway Endpoints and Interface Endpoints.
For example, an EC2 instance in a private subnet can access Amazon S3 using an S3 Gateway Endpoint without requiring a NAT Gateway for that S3 traffic.
Example:
Private EC2
↓
VPC Endpoint
↓
S3
Simple ga gurthupettuko:
VPC Endpoint = **Private access to supported AWS services**

---

20. What is a VPC Peering vs VPC Endpoint?
Answer:
VPC Peering is used to connect one VPC with another VPC.
A VPC Endpoint is used to privately access supported AWS services from a VPC.
So they solve different problems.
For example:
VPC-A
↓
VPC Peering
↓
VPC-B

Whereas:
VPC
↓
VPC Endpoint
↓
S3
Simple ga gurthupettuko:
VPC Peering = **VPC ↔ VPC**
VPC Endpoint = **VPC ↔ AWS Service**

---

21. What is a Default VPC?
Answer:
A Default VPC is a VPC that AWS creates automatically in supported Regions for an AWS account.
It comes with default networking resources that make it easier to launch resources quickly without creating the entire network setup manually.
For example, when starting with AWS, we can launch an EC2 instance in the Default VPC without first creating a custom VPC.
Example:
AWS Region
↓
Default VPC
↓
Default Subnets
↓
EC2
Simple ga gurthupettuko:
Default VPC = **AWS-provided starting network**

---

22. What is a Custom VPC?
Answer:
A Custom VPC is a VPC that we create based on our application requirements.
We decide the CIDR range, subnets, routing, internet access, and security design.
Custom VPCs are commonly used for production applications because they give us more control over the network.
For example, we can create separate public and private subnets for an application.
Example:
Custom VPC
|
├— Public Subnet
| ↓
| Load Balancer
|
└— Private Subnet
↓
EC2
↓
RDS
Simple ga gurthupettuko:
Custom VPC = **Our own network design**

---

23. What is VPC Flow Logs?
Answer:
VPC Flow Logs capture information about network traffic going to and from network interfaces in the VPC.
They are useful for troubleshooting connectivity problems and understanding which traffic is being accepted or rejected.
Flow Logs can be published to supported destinations such as CloudWatch Logs or Amazon S3.
For example, if an EC2 instance cannot connect to another resource, Flow Logs can help us investigate the network traffic.
Example:
EC2
↓
Network Traffic
↓
VPC Flow Logs
↓
Logs
↓
Troubleshooting
Simple ga gurthupettuko:
Flow Logs = **Network traffic information**

---

24. What is a Route Table Association?
Answer:
A route table association connects a subnet with a route table.
This tells the subnet which routing rules should be used for its network traffic.

A subnet can be associated with a route table, and multiple subnets can use the same route table when required.
For example, we can associate a public subnet with a route table that has a route to an Internet Gateway.
Example:
Public Subnet
↓
Route Table
↓
Internet Gateway
Simple ga gurthupettuko:
Route Table Association = **Subnet ki routing rules connect cheyyadam**

---

25. What is a NAT Instance?
Answer:
A NAT Instance is an EC2 instance configured to provide network address translation for resources in private subnets.
It was an older approach compared with the managed NAT Gateway service. A NAT Instance requires us to manage the EC2 instance ourselves.
For example, a private EC2 instance can send outbound traffic through a NAT Instance to reach the internet.
Example:
Private EC2
↓
NAT Instance
↓
Internet Gateway
↓
Internet
Simple ga gurthupettuko:
NAT Instance = **EC2-based NAT**

---

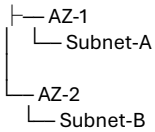
26. What is the difference between NAT Gateway and NAT Instance?
Answer:
A NAT Gateway is an AWS managed service, while a NAT Instance is an EC2 instance that we manage ourselves.
NAT Gateway generally requires less management, while NAT Instance gives more control but requires us to handle the server.
For example:
Private EC2
↓
NAT Gateway
↓
Internet
or:
Private EC2
↓
NAT Instance
↓
Internet
Simple ga gurthupettuko:
NAT Gateway = **Managed**
NAT Instance = **Self-managed EC2**

---

27. What is the difference between Region and Availability Zone?
Answer:
A Region is a separate AWS geographic area that contains multiple Availability Zones.
An Availability Zone is an isolated location inside a Region.
We use multiple AZs to improve availability and reduce the impact of a failure in one AZ.
For example:
AWS Region
|
├— AZ-1
├— AZ-2
└— AZ-3
Simple ga gurthupettuko:
Region = **Big AWS location**
AZ = **Separate location inside Region**

---

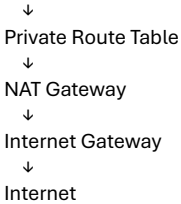
28. Can two subnets be in different Availability Zones?
Answer:
Yes. We can create different subnets in different Availability Zones within the same Region.
This is commonly done to spread application resources across multiple AZs and improve availability.
For example, an application can have one subnet in AZ-1 and another subnet in AZ-2.
Example:
VPC
|



Simple ga gurthupettuko:
Multiple Subnets + Multiple AZs = **Better availability**

---

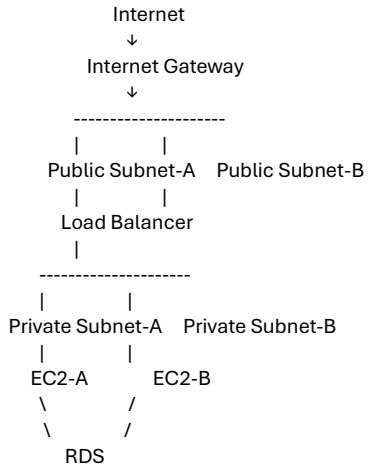
29. What happens when a resource in a private subnet needs internet access?
Answer:
A resource in a private subnet can use a NAT Gateway for outbound internet access.
The private subnet route table sends internet-bound traffic to the NAT Gateway. The NAT Gateway is placed in a public subnet and uses an Internet Gateway for internet connectivity.
For example, a private EC2 instance needs to download a package from the internet.
Example:



Simple ga gurthupettuko:
Private EC2 → **NAT Gateway → Internet**

---

30. Explain a complete VPC architecture.
Answer:
A common production VPC architecture has public and private subnets across multiple Availability Zones.
Public subnets can contain internet-facing resources such as Load Balancers. Private subnets can contain application servers and databases. Route Tables control traffic, Internet Gateway provides internet access, NAT Gateway provides outbound internet access for private resources, and Security Groups and NACLs control network traffic.
For example, users access a Load Balancer, the Load Balancer sends requests to private EC2 instances, and the EC2 instances connect to an RDS database.
Example:



Simple ga gurthupettuko:
VPC = **Network**
Subnet = **Small network**
Route Table = **Traffic direction**
Internet Gateway = **Internet connection**
NAT Gateway = **Private outbound internet**
Security Group = **Resource firewall**
NACL = **Subnet firewall**
VPC Endpoint = **Private AWS service access**